

Visual Evidence-Aware for Object Hallucinations Rectification in LLM-Based Video Captioning

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Abstract—Recent neural models for video captioning are typically built using a framework that combines a pre-trained visual encoder with a large language model (LLM) decoder. However, large language models in video captioning often generate non-existent entities, known as object hallucinations, which severely limit performance. To mitigate object hallucinations, two key issues remain: 1. Biased training data and Knowledge bias in LLM leads models to generate hallucinations; 2. Current methods focus on removal rather than restoring the correct visual content, reducing caption completeness. To address these issues, we propose a visual evidence-aware for object hallucination rectification in LLM-based video captioning. Generally, our model aims to diagnose and correct those generated object hallucinations, and then supplement missing visual content by constraining the process of text description generation. Specifically, we first generate captions by words based on the input video. When decoding each object description, the decoder utilizes visual features for hallucination diagnosis and correction, proposing visual evidence to modify hallucinatory descriptions. This process ensures the generated captions align with the visual content, alleviating the generation of object hallucinations. Compared with the baseline models, our method performs state-of-the-art performance in video captioning, especially avoiding neglecting objects in the visual content caused by the generated hallucinatory descriptions.

Index Terms—Video captioning, object hallucination, hallucination rectification, visual evidence.

I. INTRODUCTION

VIDEO captioning is the task of generating textual descriptions or subtitles for video content, aiming to express the objects, actions, and other content shown in the video through natural language [1], [2], [3], [4], [5], [6]. Video captions can enhance the understanding of video content, enabling users to comprehend the video content more easily without relying solely on watching the video itself. Additionally, accurate video captioning can assist visually impaired and hearing-impaired individuals

in understanding video content, improving their audiovisual experience and information acquisition capabilities [7], [8].

Recent advancements in multimodal large language models (MLLM) have enabled computers to understand visual content to obtain textual outputs [9]. Despite the performance improvements of these models, there are instances where the generated captions do not accurately align with the input visual content. This phenomenon is referred to as the “visual hallucination” problem [10], [11]. Specifically, visual hallucination refers to the presence of information in textual descriptions that do not exist in the visual input. In video captioning, three main types of visual hallucinations are identified: object hallucination, relation hallucination and action hallucination [12]. Specifically, object hallucination is the appearance of irrelevant objects in the descriptive text generated by the model, which is related to the model’s incorrect object categories. Since the object objects in the descriptions are often a key element of the content, resolving object hallucinations is crucial for generating accurate descriptions.

To solve the visual hallucinations, there are two mainstream approaches: adjusting training data and optimizing model methods. The former involves constructing more balanced and comprehensive training data to avoid hallucinations caused by data biases [13], while the latter focuses on mitigating hallucinations through model design [12], [14] and post-hoc correction methods [15], [16], [17], [18]. As shown in Fig. 1, we compare the results of existing visual content description methods with two hallucination mitigation approaches. We present descriptions generated by the baseline model with hallucinations [19], text generated using model optimization methods [14], and text generated by training models with balanced training data [13]. In the descriptions generated by the baseline model, we observe visually irrelevant words such as “surfboard”, which is identified as a hallucinatory word related to objects in the visual scene, such as “beach”. The reason for this hallucination is attributed to biases learned by the model from the training data. In reality, “surfboard” often co-occurs with “beach” in authentic data, whereas “clock” is unlikely to appear in this context. We can observe that balancing the co-occurrence frequency of objects in the training data is an effective solution [13], which also demonstrates the close relationship between data bias and hallucination generation. However, this method still has significant limitations. Firstly, it is infeasible to collect any possible co-occurrence video-text object pairs in training data. Further, video data contains objects with a considerable amount of repetitive information

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Hallucinated captions: A man standing on a beach **holding a surfboard**.
 Model optimization methods: A man standing on top of a sandy beach.
 Balancing training data: A man standing on a beach **with a clock**.

Fig. 1. The comparison between visual description texts with object hallucinations and the results after applying two hallucination mitigation methods.

across consecutive video frames, making it equally challenging to balance the occurrence frequency of objects in video data. Therefore, we explore how to mitigate the impact of this object data bias on video captioning tasks from a modeling perspective.

For humans, such knowledge biases do not lead to incorrect descriptions empirically, as humans only focus on the current visual content when describing visual information. Specifically, humans describe the visual content they see accurately when creating captions, without adding unrelated but commonly seen text casually. However, existing visual text generation models, based on architectures like RNNs and Transformers, employ probability prediction to iteratively generate text [4], [5]. The probabilities of the generated results are influenced by both the model's knowledge and the visual content, leading to hallucinations when there are conflicts between the two. Thus, one approach to addressing the visual hallucination problem is to use visual constraints to reduce the influence of irrelevant knowledge on the generation process.

Additionally, we observe that in some generated descriptions, object hallucinations can obscure important visual content. For example, in a hallucinatory description where the term “surfboard” appears, we can see that its presence replaces the original “clock” from the visual content. Although increasing the co-occurrence frequencies of those uncommon appearing pairs in training data is an effective solution [13], it is challenging to implement this approach on existing large-scale datasets or more complex video data. Existing model-based hallucination text correction methods tend to remove hallucinatory descriptions from the generated text, struggling to restore the obscured actual visual information into the modified text descriptions [14]. Some methods effectively reduce the degree of hallucination in description sentences, but the accuracy and completeness of the description text are still limited.

In summary, the main reasons for object hallucinations in video captioning include two points: (1) Biased training data: Objects that frequently appear in visual instructions or co-occur are more prone to causing hallucinations in the model [17]. (2) Knowledge bias in large language models: Firstly, the current approaches still use Maximum Likelihood Estimation (MLE) as the objective function, which blindly fits label data, leading to reasoning errors consistent with LLM [14]. Secondly, the knowledge aggregation mechanism of LLM causes excessive reliance on summary tokens for subsequent token generation, neglecting detailed exact visual tokens, and generating hallucinations [20].

In existing hallucination mitigation approaches within the video/image captioning domain, the dominant strategy involves adjusting training data or modifying training methodologies. However, under the current trend of widespread adoption of large-scale pre-trained models, such strategies are undoubtedly costly. Moreover, even if the training data is balanced, discrepancies still exist between its distribution and the real-world open-domain data distribution.

Therefore, we propose a post-hoc correction approach. Since visual hallucinations refer to object or action descriptions in the generated text that are inconsistent with the visual content, we argue that real-time constraints based on visual information can effectively alleviate hallucinations by ensuring that only content grounded in visual input is generated. Importantly, this constraint mechanism is agnostic to the specific vision-language architecture (i.e., the visual encoder and text decoder), allowing it to be plugged into most existing vision-language models. With only minimal training cost, we train a hallucination diagnosis and correction module, without needing to fine-tune the entire model—demonstrating both the generalizability and applicability of the approach. Specifically, our method leverages actual visual information to mitigate errors introduced by statistical co-occurrence biases. During the generation process, the model reintroduces visual signals and jointly utilizes both visual and linguistic modalities to construct knowledge-based constraints. These constraints guide the decoding process, enabling the model to correct hallucinated descriptions and recover missing visual elements, ultimately improving the alignment between generated text and the visual input. Our framework consists of a video captioning generation module and an object hallucination rectification module. The former module is responsible for generating textual descriptions of videos using visual information, while the object hallucination rectification module performs the diagnosis of hallucinations and corrects hallucinatory words timely.

Furthermore, current video caption hallucination evaluation methods are inadequate in accurately assessing the quality of generated sentences. Inspired by hallucination evaluation methods in natural language processing, we attempt to use the GPT-3.5-Turbo model as a hallucination evaluation tool. By designing prompts, we enable the model to simultaneously consider three aspects: hallucination level, sentence quality, and information completeness to evaluate the overall quality of the captions. Our contributions can be summarized as follows:

- We propose a novel visual evidence-aware model for object hallucination rectification in LLM-based video captioning, aiming to reduce object hallucinations and improve performance in video description generation.
- To accurately diagnose object hallucinations, we introduce visual feature constraints during the text generation process to determine the presence and type of hallucinations. Furthermore, to correct object hallucinations, we design visual object evidence to make real-time adjustments to descriptions diagnosed as hallucinations, thus avoiding hallucination generation.
- We propose a novel hallucination evaluation method that provides a more comprehensive assessment of video caption quality. Extensive evaluations demonstrate the effectiveness of our approach with significant improvements, alleviating the hallucinations in video captioning.

II. RELATED WORK

A. Video Captioning

Video captioning is to generate accurate and descriptive textual descriptions for videos, enabling a comprehensive and precise understanding of their content. Video captioning methods currently leverage transformer architectures as the backbones, which dramatically improve the performance [4], [5], [21]. For example, SMAN [22] utilizes additional visual and textual historical information as contextual features and employs a stacked multimodal attention network to progressively process different features, further improving the generated results. Another important research direction focuses on improving video captioning by leveraging more accurate and fine-grained visual representations. For instance, HSRA [23] introduces a three-level semantic structure of object-action-event to explicitly model multi-granular visual semantics. It further enhances the representation of action semantics by incorporating learnable action queries based on DETR. CVG [24] proposes two branches—Spatial-temporal Entity Grounding and Action Grounding—to anchor entities and actions in both spatial and temporal dimensions, respectively. EvCap [25] leverages the element extraction approach from template-based methods to reinforce the model's perception of specific elements by incorporating linguistic features in addition to general visual features. Additionally, universal vision and language pre-trained models have also driven the development of video captioning [26], [27]. MPLUG-Owl2 designs a modular network, integrating shared functional modules to facilitate modality collaboration [27]. These large-scale models are pre-trained on extensive visual-text pairs to learn universal multimodal features, which can be fine-tuned for downstream tasks such as video captioning. Despite achieving high-quality video caption generation, these methods still suffer from hallucinations, significantly limiting the application of video captioning.

B. Hallucinations in Visual-Language Models

Hallucinations in Visual-Language Models, abbreviated as visual hallucinations [16], [28], refer to incorrect or nonexistent information appearing in textual descriptions that do not

correspond to the visual input. Object hallucinations directly impact the reliability of video description models, thus garnering widespread attention from researchers. Bitten AF employs a sampling strategy to artificially alter objects in captions, balancing the co-occurrence frequency of objects in the training data [13]. This method reduces the model's reliance on linguistic priors and visual features. Similarly, LRV-Instruction mitigates hallucinations by limiting the length of generated text [10]. Besides, THVC employs a training method that extracts and interacts with features such as actions and objects in visual content separately to reduce hallucinations [12]. This kind of approach is based on the observation that frequent occurrences of actions and objects in visual instructions can mutually influence each other to produce hallucinations. Additionally, Woodpecker verifies objects in generated text using visual information and designs corresponding cue words to modify hallucinatory expressions via a language model [15]. Further, OPERA attributes hallucinations to models overly trusting key summary words already generated, neglecting visual information [20]. To address hallucinations, OPERA proposes a decoding method based on excessive trust penalties and retrospective allocation strategies in Multimodal large language models (MLLM). However, these methods still cannot avoid the issue of data bias inherent in pre-trained models leading to hallucinations. Additionally, due to the lack of comprehensive visual information constraints in the methods for adjusting hallucinations, the replaced visual information cannot be recovered by the hallucinatory words.

C. Retrieval Augmented

Retrieval Augmented is used in information retrieval tasks where additional information is introduced to improve retrieval performance [29]. Due to inherent limitations in training data and data biases, language models are increasingly being augmented with additional knowledge from internet-derived datasets to enhance their knowledge base, as evidenced by prior work [30], [31], [32]. Previous studies have shown that methods incorporating external knowledge retrieval can effectively mitigate hallucination phenomena in generated text [33]. Our approach shares common ground with these methods by using information related to a given video to correct generated hallucinatory text. However, applying retrieval augmentation methods to the visual-language domain is challenging. Traditional retrieval augmentation methods typically only handle textual data and retrieve relevant knowledge through text-based retrieval, which is unsuitable for crossmodal data like video-text pairs. To address these critical differences, we propose a strategy for constructing a structured visual knowledge base based on videos and queries. This strategy leverages scene graph knowledge from video understanding to assist in implementing cross-modal retrieval augmentation methods, mitigating hallucination issues in video captioning.

III. METHODOLOGY

We propose a visual evidence-aware model for object hallucination rectification in LLM-based video captioning. The overview of our proposed framework is illustrated in Fig. 2.

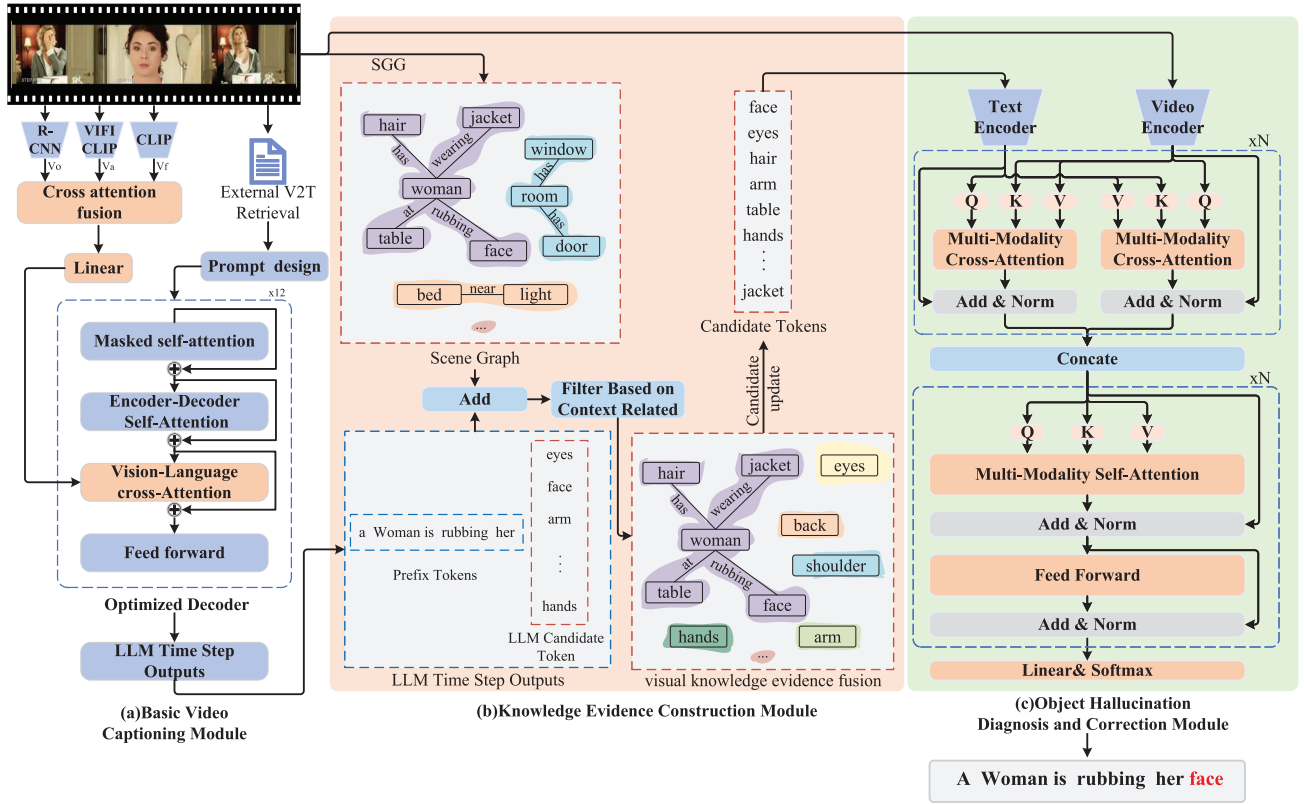


Fig. 2. Overall framework of our approach. Given a video and an external text database, our approach consists of three main modules: (a) Basic video captioning module, which generates corresponding text descriptions based on the input video and text data; (b) Knowledge evidence construction module, which constructs visual knowledge evidence using visual information and the generated captions; (c) Object hallucination diagnosis and correction module, which determines whether the current generated description contains hallucinations. Subsequently, we generate visual knowledge from the original video and use it as evidence to correct any hallucinations in the generated descriptions.

The components of our framework consist of: (a) Basic Video Captioning Model. This module comprises two main parts, including a video encoder responsible for extracting visual features from input videos, and a text decoder composed of a pre-trained language model. These two parts are integrated using a cross-attention mechanism to fuse visual features into the language model and generate textual captions. (b) Knowledge Evidence Construction Module, which constructs visual knowledge evidence using visual information and the generated captions. (c) Object Hallucination Diagnosis and Correction Module, which determines whether the current generated description contains hallucinations. Subsequently, we generate visual knowledge from the original video and use it as evidence to correct any hallucinations in the generated descriptions. The module (a) and (c) are trained separately using different loss functions. In the following sections, we will elaborate on each step in detail.

A. Basic Video Captioning Module

First, we construct a video captioning model based on the methodology of SmallCap [34]. We consider the differences between videos and images and apply this framework to video description tasks. Our basic model framework integrates a powerful pre-trained model into an encoder-decoder architecture, as illustrated in Fig. 2 (a). For the encoder, we employ three methods to extract global, object, and action features

from the video, respectively, and fuse these features into a comprehensive video representation. We utilize CLIP [35] to extract global video features, which generates visual embeddings $F = f_1, f_2, \dots, f_n$ for a sequence of video frames. Then, we remove redundant video frames by calculating cosine similarity between all video frame embeddings, resulting in a video feature V_f composed of embeddings from key video frames. As described in equation (1), we initialize the keyframe v_i selection with the first frame. For each subsequent frame, we calculate its cosine similarity with the most recently selected keyframe. If the similarity falls below a predefined threshold, the frame is selected as the next keyframe. This process continues iteratively to construct the sequence of keyframe embeddings. Here, $\text{RemoveRedundancy}(\cdot)$ denotes the removal of redundant features based on cosine similarity, where v_i represents the index positions of the preserved features in F after removing redundancy.

$$\begin{aligned} V_f &= \text{RemoveRedundancy}(f_{v_1}, \dots, f_{v_2}, \dots, f_{v_i}, \dots, f_n) \\ &= f_{v_1}, f_{v_2}, \dots, f_{v_i} \end{aligned} \quad (1)$$

Next, we employ Faster R-CNN [36] to extract features of the top K confident objects $V_o = o_1, o_2, \dots, o_k$ from video frames. As shown in equation (2), inspired by the VIFI-CLIP [37] approach, we capture motion information in the video by averaging the pooled vectors of CLIP features from adjacent h frames around key video frames, obtaining action

features $V_a = a_1, a_2, \dots, a_i$. This strategy allows us to model temporal dynamics within short video segments. The feature representation of an individual motion segment is formally defined in equation (2).

$$a_i = \text{AvgPooling}(f_{v_i-h}, \dots, f_{v_i}, \dots, f_{v_i+h}) \quad (2)$$

Finally, the three features are concatenated and fed into a Transformer, where the self-attention mechanism is employed to fuse them and obtain the final visual feature V . Next, we apply another linear layer to map the features to the same dimensionality as the decoder's hidden space. Here, $[:, :]$ denotes the operation of feature concatenation.

$$V = \text{Linear}(\text{Transformer}([V_f; V_o; V_a])) \quad (3)$$

We employ GPT-2 [38] as the decoder. Similar to SmallCap [34], since the two models operate in separate latent spaces, we connect them using multi-head cross-attention, where each layer of the decoder attends to the outputs of the encoder. To reduce computational demands during training and maintain the generalization ability of the encoder and decoder components, we froze the encoder and decoder and only trained the randomly initialized cross-attention layers between them. Similar to other retrieval-augmented models, we leverage external textual knowledge to enhance the generation capability of our model. Specifically, here we utilize the full CLIP model with its visual and textual encoders to map keyframes from videos and textual descriptions into a shared vector space. We employ the nearest neighbor search based on cosine similarity to retrieve k text items most similar to the video content from the data repository, as shown in equation (4). First, we compute the cosine distance between the video frame features and textual features, selecting the top k text descriptions based on the sorted distances.

$$s(v_i, t_j) = \frac{w_j^T \hat{z}_i}{\|w_j\| \|\hat{z}_i\|}, \quad i = 1, 2, \dots, M, \quad j = 1, 2, \dots, N \quad (4)$$

Here, $\hat{z}_i$ represents the video frame feature, w_j indicates the textual features from text t_j extracted by the CLIP text encoder, and $s(v_i, t_j)$ represents the cosine similarity between the video v_i and textual t_j . M denotes the total number of video features, and N denotes the total number of textual features. Finally, the top K textual descriptions with the highest similarity scores are extracted from the video retrieval results.

The retrieved text is concatenated into a simple prompt text, aiming to provide the model with other similar video-image text descriptions as References. The decoder takes this prompt as input and generates captions based on the image features V . The weights in the cross-attention layer are trained by minimizing the cross-entropy loss between the reference captions and the predicted tokens.

B. Knowledge Evidence Construction Module

As shown in Fig. 2 (b), our approach performs hallucination diagnosis and correction at each time step of the LLM decoder, with visual knowledge evidence being crucial for hallucination correction. Therefore, at every time step where a hallucination occurs, we need to construct new visual knowledge evidence

based on both visual and contextual information. First, at each decoding time step of the LLM, we retrieve the current output, the previously generated sentence prefix (Prefix Tokens) and the set of candidate words generated at the current step (LLM Candidate Tokens). Through statistical analysis of the output candidate words, we observed that when the highest-probability candidate word sequence contains hallucinations, selecting alternative words with slightly lower probabilities can reduce the likelihood of hallucinations in the overall sentence. Therefore, we retain a subset of high-probability candidate words as the final candidates for hallucination correction.

Videos and generated text belong to different modalities, making it challenging to directly use key visual content as evidence to modify hallucinatory words. Our approach involves directly converting key visual information into structured textual knowledge evidence. Here, we employ scene graph generation methods from visual understanding [39], which effectively extract object and relationship information from visual data. Specifically, we input images extracted from the previously obtained video keyframe sequence into a trained Scene Graph Generation (SGG) model to generate scene graphs. We select the top m_1 object entities with the highest confidence scores and combine them with m_2 relationship pairs to form structured key knowledge $R\{r_1, \dots, r_m\}$ for the video.

Although structured video knowledge contains comprehensive visual content, it still introduces a significant amount of noise for correcting hallucinations in subtitles. Therefore, we use the prefix of the generated sentence as a basis to filter the visual information. Specifically, we extract all the key terms $C\{c_1, \dots, c_n\}$ from the generated text and compute their semantic distance to each element in the structured video knowledge.

$$S(R, C) = \sum_{i=1}^m \frac{1}{n} \sum_{j=1}^n \cos(f(r_i), f(c_j)) \quad (5)$$

Based on these semantic distances, we select the top M closest elements as the visual knowledge evidence. Finally, we combine this visual knowledge evidence with the updated LLM Candidate Tokens to form the final hallucination diagnosis knowledge evidence.

C. Object Hallucination Diagnosis and Correction Module

1) *Video Caption Hallucination Detection Dataset Construction*: Given the absence of existing datasets for word-level hallucination detection in the visual description text, we create a new video captioning hallucination detection dataset based on existing video captioning datasets. This dataset contains fine-grained annotations to identify whether the descriptions generated by a video captioning model are hallucinations. The dataset consists of video-caption pairs sampled from the training sets of the MSVD and MSR-VTT datasets, which are manually annotated to form two hallucination validation datasets: MSVD Hallucination Detection Dataset (MSVD_HD) and MSR-VTT Hallucination Detection Dataset (MSR-VTT_HD). The goal of this dataset is to enable the model to learn to diagnose whether words generated by a video captioning model are hallucinations. Therefore, it is



text_hallucination: a long haired attractive woman places fried **chips**
text_label: a long haired attractive woman places fried **chicken**
hallucination_type_text: object_hallucination
ground_truth: a long haired attractive woman places fried chicken on to a plate

Fig. 3. Example of data annotation.

necessary to manually identify the positions of hallucinatory words in the generated sentences, and then annotate them along with the sentence prefixes according to the corresponding hallucination types. The annotation process categorizes whether hallucinations occur and their types into three classes: (1) No hallucination, (2) Object hallucination, and (3) Action hallucination. We define these categories as follows:

- **No Hallucination:** Objects or actions exist in the video, and their descriptions are accurate based on the video. Any described relationships can be accurately inferred from the video.
- **Object Hallucination:** The words in the current generated description refer to objects that do not exist in the video or whose descriptions are inaccurate.
- **Action Hallucination:** The words in the current generated description refer to actions or relationships that do not exist in the video or whose descriptions are inaccurate.

We provide detailed annotations for these three categories based on the detailed descriptions corresponding to the videos. The descriptions are provided at the clause level, but the annotations are labeled based on the final words. Specifically, each annotation represents the hallucination type of the last word generated in the current clause, as illustrated in Fig. 3.

Since the decoding process of the language model generates words by predicting the next word iteratively, our model needs to immediately assess whether the currently generated word is a hallucination as each word is generated. Therefore, our dataset only needs to capture the clauses where the last word of the generated text is identified as a hallucination, without the necessity of generating a full description.

To reduce the cost of annotating hallucinated words, we construct a new dataset using existing video-text descriptions. We use all video-text pairs in the training set of the existing video description dataset as raw data. First, we truncate each label text in the dataset (randomly cutting after a verb or noun in the sentence, and retaining only the earlier part of the sentence). We ensure that the number of object and action words are balanced. Next, we randomly replace the last object or action word in the truncated clause with a non-existent word in the video, thereby creating a clause containing an object or action hallucination. The specific replacement strategy is as follows: for objects, we choose from high-frequency words or words with high co-occurrence with the object in the current clause based on the dataset's word statistics. To ensure

TABLE I
STATISTICS OF HALLUCINATION DIAGNOSIS DATASET

Dataset	Video	Video-Text	Train	Val	Test
MSVD_HD	1200	47007	37539	4817	4651
MSR-VTT_HD	6500	123584	98740	12303	12541

grammatical correctness, the final replacement is refined using a language model.

Based on the statistical analysis of hallucination types observed in the generated sentences, we determine the final proportions for the three annotation types: 30% without hallucinations (with 20% for no object hallucinations and 10% for no action hallucinations), 35% for object hallucinations, and 35% for action hallucinations. The dataset statistics are shown in Table I.

2) *Diagnosis Part:* We implement a hallucination diagnosis model to detect whether the words generated by the video captioning model are hallucinations. As shown in Fig. 2 (c), we design a diagnosis model that mirrors the architecture of the video captioning model. This design choice enhances training robustness and improves the quality of sentence replacements after hallucination detection. The diagnosis model consists of an encoder that extracts both video and textual features. A cross-attention module is employed to align and model the interaction between video and text features, where an additional modality-assisted alignment is introduced to capture the matching information between the two features via attention computation. Furthermore, we utilize a self-attention module between the video-text features, enhanced through cross-modality interaction, to better model multimodal information. This approach aims to strengthen the alignment between modalities and capture unique feature information from each modality. Finally, a linear classification head is added to diagnose hallucination categories.

We train the diagnostic model at the word-level granularity. Specifically, the model utilizes information from the video and the context of the caption to determine if the last word in the caption contains a hallucination. For the labeling of hallucination data in the dataset, we first map the hallucination categories to traditional one-hot encoded labels. To prevent this vector format from adversely affecting the model's performance, and to ensure that the hallucination labels accurately reflect the degree of hallucination in the current sentence, we draw inspiration from label smoothing techniques. We map the data labels to values between 0 and 1 by utilizing the similarity between visual and textual features. However, unlike traditional approaches, our mapping is closely tied to the data and represents meaningful pseudo-label annotations, as shown in the following equation:

$$y = \begin{cases} (1 - \epsilon)y_i + \epsilon \cdot \cos(V, T) \cdot \tau, & y_i \in R_1 \\ (1 - \epsilon)y_i + \frac{\epsilon \cdot (1 - \cos(V, T) \cdot \tau)}{C - 1}, & y_i \in R_2 \end{cases} \quad (6)$$

In this context, ϵ represents the smoothing parameter that controls the degree of label smoothing, C denotes the number of categories, and y_i is the i -th element in the original label.

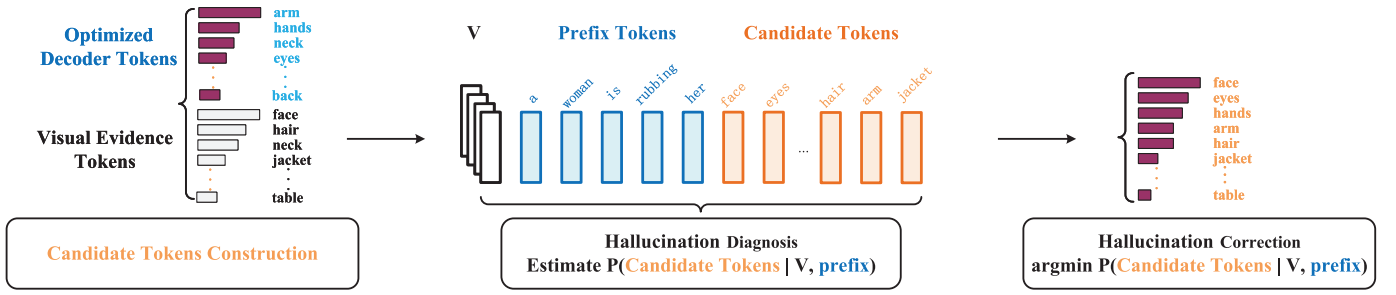


Fig. 4. The process of hallucination correction.

V and T refer to the video and textual features encoded by CLIP, respectively, while $Cos(\cdot)$ computes the cosine similarity between two vectors. R_1 represents all the label categories without hallucination, and R_2 represents all the label categories with hallucination. Additionally, τ is a hyper-parameter used to adjust the similarity weight. For each video-text pair, we perform a forward pass similar to video captioning. At the final embedding layer, we map to target class labels using the classification head. We optimize using cross-entropy loss.

3) *Correction Part*: Under the guidance of visual knowledge evidence, the hallucination diagnosis model can also serve as a corrector to modify hallucinations in the generated output. As shown in Fig. 4, upon diagnosing the current generated word as a hallucination, we combine structured visual information with the original candidate words based on the type of hallucination. Using the hallucination diagnosis model, we identify and directly replace the word with the lowest likelihood of hallucination as the current generated word. This timely guidance directs the video captioning model to focus attention on video content while reducing the probability of generating subsequent hallucinatory words.

$$x_t = \arg \min_{w \in W^{(k)}} H(w | V, x_{t-1}, W^{(k)}) \quad (7)$$

As shown in equation (7), $W^{(k)}$ represents a set consisting of the top- n predicted results of the probability distribution from the video captioning model, corresponding to the relevant portion of the structured visual evidence (for object hallucinations, extracting relevant object candidate information from the video evidence; for action hallucinations, extracting relationships and action information from the visual content). x_{t-1} denotes the previously generated text prefix. V represents the complete video features. H denotes the probability of hallucination existing in the current sequence, which is calculated by using the hallucination diagnosis model given the video features and text prefixes. x_t represents the word output chosen at the current time step.

IV. EXPERIMENT

A. Datasets

The Microsoft Video Description Corpus (MSVD) dataset is a benchmark dataset for video captioning tasks collected by Microsoft Research Asia [40]. This dataset consists of 1970 short video clips along with corresponding natural language descriptions, covering various scenes and actions. Each video

segment lasts from a few seconds to several tens of seconds and is accompanied by human-generated descriptions to aid in understanding the video content. For each video, the dataset includes 40 captions generated by Amazon Mechanical Turk. To ensure a fair comparison, we follow standard splits [41], where 1200 videos are used for training, 100 for validation, and 670 for testing.

The Microsoft Research Video-to-Text (MSRVTT) dataset is developed by Microsoft Research to advance research in video content understanding and natural language description generation [42]. This comprehensive dataset comprises approximately 10,000 video clips sourced from real-world video websites, with an average length of 20 seconds per clip. Each clip is associated with 20 text descriptions provided by different individuals, totaling around 200,000 descriptions. These descriptions are generated by Amazon Mechanical Turk workers to ensure quality and diversity. The MSRVTT dataset offers consistent video resolutions and can be used for various video understanding and generation tasks, such as content description, video summarization, and video retrieval. We adhere to standard benchmark splits [42], where 6513 samples are used for training, 497 for validation, and 2990 for testing purposes.

Hallucination Detection Dataset, as introduced in Section III-C above. This dataset comprises video-caption pairs sampled from the training sets of MSVD and MSRVTT datasets. The MSVD Hallucination Detection Dataset is divided into a training set with 37,539 video-text pairs and a test set with 4,651 video-text pairs. The MSRVTT Hallucination Detection Dataset consists of a training set with 98,740 video-text pairs and a testing set with 12,541 video-text pairs.

B. Implementation Details

The encoder and decoder of the base video captioning model are initialized from CLIP-VIT-L/14 and GPT-2Base, respectively. The encoder and decoder parameters are fixed, and only the cross-attention layers between them are trained. The number of encoded video keyframes is dynamically determined by the model, with a maximum of 80 frames in the current dataset. The Faster R-CNN in the encoder extracts features from up to 15 objects. Within the 12 layers of GPT-2, each layer includes an additional 12-headed cross-attention layer. The cross-attention layers are trained separately on the MSVD and MSRVTT datasets using standard splits. The AdamW optimizer is used to minimize the cross-entropy

loss, with an initial learning rate of $1e-4$ and a batch size of 64. Training runs for 20 epochs, and we use the checkpoint with the best CIDEr score on the validation set.

The design of the hallucination diagnosis model is similar to the base video captioning model. We initialize the video encoder from CLIP-VIT-L/14 and employ the same encoding strategy. For the decoder, we use a simplified two-layer Transformer as the decoder, followed by a fully connected network and softmax layer to map the decoder outputs to hallucination categories. We optimize the model using the cross-entropy loss function. The batch size is set to 20, and we train for 20 epochs on a single training dataset, selecting the model with the highest classification accuracy on the validation set. In the hallucination correction validation strategy, we compile candidate lists consisting of the top 100 tokens from the language model candidates and the top 100 relationships from the visual constraint information.

C. Evaluation Metrics

To evaluate the performance of our model, we use the COCO evaluation toolkit to compute standard metrics: BLEU-4 (B@4) [43], METEOR [44], CIDEr [45], and SPICE (S) [46]. Additionally, to provide a more precise assessment of hallucination levels within the video caption content, we utilize the semantic distance measured by COAHA [12] as a hallucination evaluation metric, where OH and AH represent Object Hallucination score and Action Hallucination score, respectively.

In addition, current video caption hallucination evaluation methods are still inadequate in assessing the quality of generated sentences. For instance, THVC evaluates the hallucination level of text by considering the average semantic distance between the objects and actions described in the generated text and those in the labels, but it overlooks the completeness of scene descriptions and the inherent quality of the text itself, both of which are crucial factors in evaluating video description quality. Thus, the current metrics do not comprehensively evaluate the quality of the sentences themselves. Inspired by the hallucination evaluation methods in natural language processing, we attempt to use the GPT-3.5 model as a hallucination evaluation tool. By designing prompts, we enable the model to consider three aspects simultaneously: hallucination level, sentence quality, and information completeness, resulting in a floating-point score ranging from 0 to 10. Hallucination level is defined as the proportion of object and action information in the sentence that does not exist in the labels. Information completeness refers to the proportion of objects and actions described in the sentence that are present in the labels. Text quality is evaluated based on the grammatical correctness and logical coherence of the description. We compared this method's scores with human ratings, and the close alignment between the two indicates the rationality of this evaluation method. The detailed prompt design is illustrated in Fig. 5.

D. Quantitative Results

Based on the model design, we first train a word-level hallucination verification model on the hallucination detection

Role Definition:
 Function: A professional text hallucination evaluation robot.
 Objective: Score the quality of evaluated text based on tag text considering hallucination, fluency, information completeness, and other aspects.
 Output: A floating-point score ranging from 0 to 10.

Input Requirements:
 Text to be Evaluated: {Generated Caption}.
 Label Text: {Label Captions}.

Evaluation Process:
Text Hallucination Level:
 Definition: Identifying descriptions (actions, objects, etc.) in the text to be evaluated that do not appear in any labeled texts.
Information Completeness:
 Definition: The degree to which key semantic information from the labeled text is included in the text to be evaluated.
Text Quality:
 Definition: Evaluating the grammar correctness and rationality of the expression in the text to be evaluated.
Scoring Criteria: Lower quality leads to a lower score, and vice versa.

Output format:
Explanation:
 Text hallucination level: {reason}.Score: {hallucination score/10}.
 Information completeness: {reason}.Score: {information score/10}.
 Text quality: {reason}.Score: {quality score/10}.
Evaluation Score:
 considering the various factors, I would score the text to be evaluated as {score} out of 10.

Fig. 5. Prompt template for GPT-3.5-Turbo evaluation.

TABLE II
HALLUCINATION DIAGNOSIS MODEL PERFORMANCE

Dataset	Method	ACC	No_ACC	Obj_ACC	Act_ACC
MSVD_HD	in-domain	0.880	0.898	0.913	0.822
	out-domain	0.860	0.902	0.836	0.850
MSR-VTT_HD	in-domain	0.823	0.888	0.806	0.786
	out-domain	0.744	0.724	0.872	0.622

dataset. The experimental results are shown in Table II, where we use “ACC” to denote the overall average classification accuracy, “No_ACC” for accuracy excluding hallucination categories, “Obj_ACC” for object hallucination classification accuracy, and “Act_ACC” for action hallucination classification accuracy, respectively. To verify the model’s generalization capability, we define “in-domain” as the model being trained solely on the current dataset, and “out-domain” as the model being trained on a different dataset without fine-tuning on the current dataset. For example, the performance of a model trained on the MSVD_HD dataset and directly tested on the MSR-VTT_HD dataset. Initially, we conduct training on MSVD_HD and directly validate the test sets of both datasets. The average classification accuracy on MSVD_HD reaches 88%, and without any additional training, we achieved a zero-shot accuracy of 74% on MSR-VTT_HD. Similarly, conducting the same experiment on the MSR-VTT_HD dataset yields overall good performance, and the cross-domain zero-shot performance on MSVD_HD surpasses our performance on the in-domain dataset. These results

TABLE III

PERFORMANCE COMPARISON ON MSVD AND MSR-VTT, RESPECTIVELY

Dataset	Method	$B@4 \uparrow$	$M \uparrow$	$C \uparrow$	$R \uparrow$
MSVD	SGN(AAAI'21) [47]	0.528	0.353	0.943	0.739
	MGRMP(ICCV'21) [48]	0.558	0.369	0.985	0.745
	THVC(ACCV'22) [12]	0.533	0.369	0.999	0.740
	VASTA(GCPR'22) [49]	0.561	0.391	1.064	0.745
	IcoCap(TMM'23) [50]	0.591	0.395	1.103	0.765
	TextKG(CVPR'23) [5]	0.608	0.385	1.052	0.751
	Blip2+GPT2(ICML'23) [51]	0.550	0.392	1.127	0.762
	LaVit+GPT2(ICLR'24) [52]	0.574	0.399	1.110	0.769
	Ours_base	0.493	0.352	0.922	0.732
	Ours	0.570	0.403	1.137	0.778
MSR-VTT	SGN(AAAI'21)	0.408	0.283	0.495	0.608
	MGRMP(ICCV'21)	0.417	0.289	0.514	0.621
	THVC(ACCV'22)	0.411	0.289	0.517	0.619
	HMN(CVPR'22) [53]	0.434	0.290	0.514	0.626
	CoCap(ICCV'23) [54]	0.444	0.303	0.572	0.634
	TextKG(CVPR'23)	0.437	0.269	0.524	0.624
	Blip2+GPT2(ICML'23)	0.404	0.291	0.537	0.614
	LaVit+GPT2(ICLR'24)	0.435	0.294	0.542	0.627
	Ours_base	0.428	0.291	0.511	0.621
	Ours	0.469	0.334	0.590	0.673

TABLE IV

ACROSS DATASETS ZERO-SHOT VIDEO CAPTIONING ON MSVD AND MSR-VTT

Dataset	Method	Pretraining Data	$B@4$	M	C	R
MSVD	Video-LLaMA(EMNLP'23) [55]	2.5Mvid+595King	0.127	0.236	0.385	0.430
	Video-LLaMA(EMNLP'23) [55]	Panda-2M	0.312	0.347	0.470	0.599
	Panda-70M-MS(CVPR'24) [56]	Panda-70M	0.328	0.353	0.492	0.612
	ours	MSR-VTT	0.439	0.296	0.550	0.630
	Video-LLaMA(EMNLP'23)	2.5Mvid+595King	0.058	0.159	0.143	0.30
MSR-VTT	Video-LLaMA(EMNLP'23)	Panda-2M	0.235	0.267	0.291	0.468
	Panda-70M-MS(CVPR'23)	Panda-70M	0.254	0.277	0.315	0.501
	ours	MSVD	0.290	0.233	0.311	0.550

demonstrate the effectiveness of our method in distinguishing hallucinatory text based on visual information, with the model exhibiting good generalization capabilities.

Table III provides a detailed comparison between the MSVD and MSR-VTT datasets. Specifically, our experimental results on the MSVD dataset indicate that our model demonstrates significant performance improvements over existing methods. Compared to TextKG [5], while B@4 favors TextKG, our model still achieves higher scores on the CIDEr metric. CIDEr is specifically designed for captioning tasks and aligns more closely with human judgment. Our method also achieves substantial improvements across all metrics on the MSR-VTT dataset, with our model's performance being noticeably superior to other methods.

Given that our experiments focused on optimizing performance improvements for tasks on specific datasets, one might question its generalizability across different datasets. Therefore, we investigated the zero-shot generalization capability of our method on different datasets. The experimental setup is as follows: we first train our method on one dataset, and then, without any fine-tuning, directly perform zero-shot video captioning on another dataset. In this experiment, we initially train our model on the MSR-VTT and MSVD datasets, respectively, and then test the model on the untrained datasets. As shown in Table IV, We observed that the results of our method are comparable to the performance of current large-scale pretrained captioning methods, and the zero-shot captioning results on MSVD achieved a significant improvement in CIDEr score.

TABLE V

HALLUCINATION PERFORMANCE COMPARISON ON MSVD AND MSR-VTT, RESPECTIVELY

Dataset	Method	$COAHA \downarrow$	$OH \downarrow$	$AH \downarrow$	$GPT3.5 \uparrow$
MSVD	THVC(ACCV'22) [12]	0.0702	-	-	-
	VASTA(GCPR'22) [49]	0.0683	0.0447	0.0235	6.415
	CoCap(ICCV'23) [54]	0.0753	0.0403	0.0248	6.277
	Blip2+GPT2	0.0714	0.0448	0.0266	6.235
	LaVit+GPT2	0.0671	0.0435	0.0236	6.551
	Ours_base	0.0801	0.0511	0.029	6.568
	Ours	0.0660	0.0441	0.0219	6.582
MSR-VTT	VASTA(GCPR'22) [49]	0.1058	0.0900	0.0158	6.558
	HMN(CVPR'22) [53]	0.0843	0.0691	0.0152	6.499
	CoCap(ICCV'23) [54]	0.0942	0.0775	0.0166	6.584
	Blip2+GPT2	0.0943	0.0778	0.0165	6.358
	LaVit+GPT2	0.0801	0.0651	0.0149	6.269
	Ours_base	0.0726	0.0570	0.0156	6.521
	Ours	0.0654	0.0510	0.0144	6.699

TABLE VI

ABLATION STUDY ON MSVD BENCHMARKS

DATASET	Method					Supervised				Hallucination		
	V2T	SGG	LLM	OBJ	ACT	B@4	M	C	R	COAHA	OH	AH
MSVD	x	x	x	x	x	0.481	0.333	0.902	0.692	0.0853	0.0522	0.0331
	✓	x	x	x	x	0.493	0.352	0.922	0.732	0.0801	0.0511	0.029
	✓	x	✓	x	x	0.500	0.377	1.001	0.745	0.0715	0.0474	0.0240
	✓	✓	x	x	✓	0.489	0.351	0.944	0.721	0.0803	0.0553	0.025
	✓	✓	x	✓	x	0.495	0.355	0.971	0.723	0.0779	0.0483	0.0296
	✓	✓	x	✓	✓	0.499	0.369	0.977	0.738	0.0728	0.0479	0.0249
	✓	✓	✓	✓	✓	0.570	0.403	1.137	0.778	0.0660	0.0441	0.0219

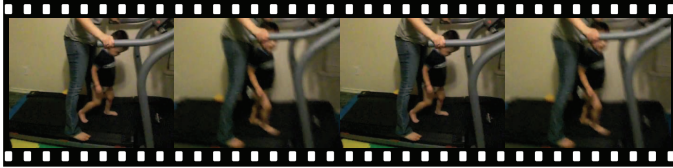
To demonstrate the effectiveness of our hallucination mitigation approach, we compared our model with state-of-the-art models specifically on hallucination generation metrics. As shown in Table V, our method reduces object hallucinations by 16% compared to the baseline THVC. Experimental results on the MSR-VTT dataset indicate that our method decreases the COAHA hallucination metric by 22% compared to HMN. The results of mitigating both types of hallucinations show that our method performs better in reducing object hallucinations in sentences. Furthermore, our approach also significantly improves traditional video captioning evaluation metrics, underscoring the substantial impact hallucinations have on video description model performance.

E. Ablation Study

We conduct quantitative evaluations to assess the effectiveness of our proposed module through ablation studies. Starting from the baseline video captioning model, we incrementally add the hallucination verification and visual information verification modules. The experimental results are presented in Table VI. Three types of hallucination constraints are primarily investigated. When none of the three modules are active in the model, it degrades to the basic video captioning model, with the hallucination detection module also removed. These constraints include (1) V2T module: Enhances the diversity of the generated text by retrieving cross-modal textual information to support the decoder.(2) SGG module: The Scene Graph Generation module serves as the foundation for constructing visual evidence. (3) Basic Hallucination Verification Modification Module (LLM), which reduces visual hallucinations by incorporating candidate tokens outputted by the decoder as constraints. (4) Visual Object Constraint, which integrates object information from visual data to generate



baseline: a **man** is mixing ingredients in a bowl
 ours: a **woman** is mixing ingredients
 ground truth: a woman is combining a mixture of mango pulp



baseline: a girl is jumping on a **chair**
 ours: a **couple** of girls are running on a **treadmill**
 ground truth: a toddler walking on a treadmill



baseline: a **sponge** is being put into a ball
 ours: a man is rubbing a **petrified animal**
 ground truth: a man is holding a frog in his hand



baseline: a girl is playing a **guitar**
 ours: a girl is playing a **violin**
 ground truth: a kid is playing violin

Fig. 6. Qualitative analysis.

visual descriptive text. (5) Visual Action/Relation Constraint, which incorporates motion information of objects and relationships between objects from visual data to generate visual descriptive text.

As shown in Table VI, introducing the V2T module alone leads to a noticeable improvement in the quality of the generated captions. However, it also results in a slight increase in hallucination instances within the descriptions. On the other hand, when the SGG module is removed, the model's ability to mitigate hallucinations significantly degrades. These results suggest that while the V2T module helps maintain the overall generation quality, the SGG module plays a crucial role in reducing hallucinations and ensuring semantic consistency. We appreciate your insightful suggestion, which helps us strengthen the empirical analysis of our proposed method. In addition, the results also demonstrate that our method is particularly effective in alleviating object hallucinations, which can be mainly attributed to the introduction of the visual grounding module.

F. Qualitative Analysis

In Fig. 6, the examples demonstrate that our proposed method effectively identifies and corrects hallucinated words generated during the descriptive process. As illustrated in the top right figure, the baseline model incorrectly identifies “frog” as “sponge.” Our method not only successfully removes the hallucinated word “sponge” but also utilizes visual evidence to restore the correct visual information related to “frog” as “animal.” Similarly, as shown in the bottom right figure, our method can also correct the erroneous object description “guitar” to the accurate description “violin” Hence, the model effectively mitigates object hallucinations. The proposed model does not merely eliminate incorrect objects; it replaces hallucinated objects with the correct choices based on visual evidence.

V. LIMITATIONS

We have identified several primary limitations of our approach, particularly regarding the fluency of the generated text compared to other methods. The analysis of error samples in the hallucination correction process revealed two main reasons for these limitations. The first reason is that replacing hallucinated text during the generation process with visual constraints impacts the current probability distribution of the language decoder, thus affecting the coherence of subsequent text generation. The second reason is that the generated visual constraint scene graphs contain multiple granularities of visual semantics, such as encompassing semantic knowledge of the same object like “tree”, “trunk” and “leaves” while the hallucination verification model is primarily equipped to assess language coherence and text hallucinations, lacking the ability to match semantic granularity in the text description. This mismatch affects text readability and consistency with reference objects in the labeled captions. To overcome this limitation, one potential solution is to incorporate semantic integrity constraints into the training process of the hallucination verification module. Another possible solution is to enhance the model's recognition ability for text fluency by designing more detailed training data.

VI. CONCLUSION

In video captioning, the presence of object hallucinations can significantly limit the accuracy and effectiveness of generated captions. To address object hallucinations in LLM-based video captioning is crucial for improving model performance. While existing research focuses on removing hallucinatory descriptions, neglecting the recovery of replaced visual content in the captions. To tackle these challenges, we propose a visual evidence-aware approach for object hallucination rectification in LLM-based video captioning. This approach involves a model that diagnoses and corrects object hallucinations while supplementing missing visual content by constraining the text

generation process with visual evidence. By utilizing a visual encoder and an LLM-based decoder with visual evidence constraints, the model generates captions that align more closely with the visual content, reducing the occurrence of object hallucinations and improving overall captioning performance. Further, our proposed method maintains the coherence between textual descriptions and visual content, avoiding the neglect of objects in the visual context.

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